Online Supplementary Material

Vilimir Yordanov*

Appendix E to “On iterated Lorenz curves with applications: the multivariate case”

Abstract: This file contains Appendix E, which is provided as supplementary material.

Keywords: Lorenz curve, iteration, contraction mapping, golden section, copula

MSC 2020: 39B12, 60F99, 62G30, 62H05, 62P05

***Corresponding author: Vilimir Yordanov,** Technical University Vienna, Financial and Actuarial Mathematics; Vienna University of Economics and Business, Vienna Graduate School of Finance, Vienna, Austria, E-mail: vilimir.yordanov@tuwien.ac.at, vilimir.yordanov@vgsf.ac.at; non-academic e-mail: villyjord@gmail.com

Appendix E

The main body of the paper works in the regular, atom-free setting. This appendix explains why this assumption is essential for the *Fréchet–Hoeffding lower-bound* comparison argument used in *Lemma 152*, and then studies the genuinely new dynamics that appear when atoms are allowed. In particular, endpoint atoms may produce valid finite iterates whose normalizing denominators converge to zero.

Throughout this appendix, atoms are treated according to the same generalized-inverse convention as in the rest of the paper. Thus an atom is not split over its rank interval[1]; it is assigned to the left edge of the corresponding marginal jump. This convention is exactly where the continuous proofs and the atomic dynamics diverge.

It is interesting to note that one-dimensional *Lorenz curve* studied in [31] is defined by the quantile integral

$$L_F(x) = \frac{\int_0^x F^{-1}(u)\,du}{\int_0^1 F^{-1}(u)\,du},$$

which automatically spreads an atom over its whole quantile interval. Hence no separate atomic case is needed in the one-dimensional theorem. The alternative threshold formula sometimes used in the economic literature

$$\frac{\int_0^{F^{-1}(x)} t\,dF(t)}{\int_0^1 t\,dF(t)}$$

coincides with the quantile formula when F is continuous, but not in general. Thus the continuous main body is consistent with [31], while the present appendix studies the additional atomic dynamics created by the generalized-inverse convention.

Failure of Lemma 152 in the atomic case

We first identify the exact point at which the proof of *Lemma 152* uses atom-freeness. Introduce for convenience the notation

$$K_n = L^n_{1-}, S_n = K_n^{-1} = T_{1-,n}, J_n = I_{-,n}.$$

In the *Fréchet–Hoeffding lower-bound* trajectory, the normalizing denominator is

$$I_{-,n} = \int\limits_0^1 T_{1-,n}(u)T_{2-,n}(1-u)\,du.$$

Under the complementary identity

$$T_{2-,n}(1-u) = 1 - S_n(u),$$

this becomes

$$I_{-,n} = \int\limits_0^1 S_n(u)(1 - S_n(u))\,du.$$

We now derive the formula for the next first marginal directly from the *Lorenz update*. The bivariate *lower-bound* iterate satisfies

$$L_-^{n+1}(x_1, x_2) = \frac{\int_0^{T_{1-,n}(x_1)} \int_0^{T_{2-,n}(x_2)} u_1 u_2\,dL_-^n(u_1, u_2)}{I_{-,n}}.$$

Setting $x_2 = 1$ gives the first marginal

$$K_{n+1}(x) = L_-^{n+1}(x, 1) = \frac{\int_0^{T_{1-,n}(x)} \int_0^1 u_1 u_2\,dL_-^n(u_1, u_2)}{I_{-,n}}.$$

1 If $K(a-) = p$ and $K(a) = p + m$, then the atom at a corresponds to the quantile plateau $(p, p + m]$, on which $K^{-1}(u) = a$. Randomizing inside the atom would spread its rank over this interval. With the generalized-inverse convention used here, no such randomization is made: the whole atom is assigned to one edge of the jump.

Equivalently, if $(Y_1, Y_2) \sim L_-^n$, then

$$K_{n+1}(x) = \frac{E\left[Y_1 Y_2 1_{\{Y_1 \le T_{1-,n}(x)\}}\right]}{E[Y_1 Y_2]}. \tag{1020}$$

Since $T_{1-,n} = S_n$, the event in the numerator is

$$\{Y_1 \le S_n(x)\}.$$

For the *Fréchet–Hoeffding lower-bound coupling* we may write

$$Y_1 = S_n(U), \qquad Y_2 = T_{2-,n}(1-U), \qquad U \sim U(0,1).$$

This gives

$$Y_1 Y_2 = S_n(U)(1 - S_n(U)),$$

and the event $\{Y_1 \le S_n(x)\}$ becomes

$$\{S_n(U) \le S_n(x)\}.$$

Substituting this into (1020), we obtain the exact generalized-inverse formula

$$K_{n+1}(x) = \frac{\int_{\{u:\, S_n(u) \le S_n(x)\}} S_n(u)(1 - S_n(u))\, du}{\int_0^1 S_n(u)(1 - S_n(u))\, du}. \tag{1021}$$

In the proof of *Lemma 152*, formula (1021) was replaced by

$$K_{n+1}(x) = \frac{\int_0^x S_n(u)(1 - S_n(u))\, du}{J_n}.$$

This replacement is valid only if

$$\{u : S_n(u) \le S_n(x)\} = [0, x] \quad \text{up to Lebesgue-null sets.} \tag{1022}$$

Equivalently,

$$K_n(S_n(x)) = x. \tag{1023}$$

Condition (1023) holds when K_n is continuous. It may fail when K_n has atoms, because then S_n has flat pieces.

In general, since S_n is nondecreasing, we have

$$\{u : S_n(u) \le S_n(x)\} = [0, \alpha_n(x)] \quad \text{up to null sets,}$$

where

$$\alpha_n(x) = K_n(S_n(x)). \tag{1024}$$

Thus the correct atomic formula is

$$K_{n+1}(x) = \frac{\int_0^{\alpha_n(x)} S_n(u)(1 - S_n(u))\, du}{J_n}, \qquad \alpha_n(x) = K_n(S_n(x)). \tag{1025}$$

Formula (840) is precisely the special case of (1025) in which $\alpha_n(x) = x$.

This is not a harmless distinction. Suppose that K_n has an atom at its upper endpoint 1, so that

$$K_n(1-) = p < 1, \qquad K_n(1) = 1.$$

Then

$$S_n(x) = 1, \qquad p < x \le 1.$$

Hence, for $p < x \le 1$,

$$\alpha_n(x) = K_n(S_n(x)) = K_n(1) = 1.$$

The correct formula (1025) gives

$$K_{n+1}(x) = \frac{\int_0^1 S_n(u)(1-S_n(u))\,du}{J_n} = 1, \qquad p < x \le 1.$$

Thus the whole upper-endpoint atom is assigned to the left edge of the jump. By contrast, formula (840) would give

$$K_{n+1}(x) = \frac{\int_0^x S_n(u)(1-S_n(u))\,du}{J_n},$$

which is generally strictly smaller than 1 for $x < 1$.

Therefore the first failure of the proof of *Lemma 152* in the atomic case is the identity (840). Consequently the next step,

$$q_n(u) = K'_{n+1}(u) = \frac{S_n(u)(1-S_n(u))}{J_n},$$

is not justified either. If K_n has an atom, then K_{n+1} may have a jump, so K_{n+1} need not be absolutely continuous. The layer-cake argument for unimodal densities leading to (855) and the sharper estimate (857) therefore does not apply. Hence *Lemma 152*, *Lemma 155*, and *Corollary 156* must be read under the atom-free hypotheses used in the main body.

The atom-free condition expressed in terms of the initial law

We now translate the previous condition into the initial law F. Let F_1, F_2 be the one-dimensional marginals of F, and write

$$Q_i = F_i^{-1}, \qquad i = 1, 2.$$

The *Fréchet–Hoeffding lower-bound* coupling associated with F_1, F_2 is

$$X_1 = Q_1(U), \qquad X_2 = Q_2(1-U), \qquad U \sim U(0,1).$$

Its initial *lower-bound* normalizer is

$$J^Q = \int_0^1 Q_1(u) Q_2(1-u)\,du. \tag{1026}$$

Assume $J^Q > 0$. The first marginal of the initial lower extremal *Lorenz curve* is

$$L^0_{1-}(x) = \frac{\int_{\{u:\, Q_1(u) \le Q_1(x)\}} Q_1(u) Q_2(1-u)\,du}{J^Q}. \tag{1027}$$

Similarly,

$$L^0_{2-}(x) = \frac{\int_{\{u:\, Q_2(1-u) \le Q_2(x)\}} Q_1(u) Q_2(1-u)\,du}{J^Q}. \tag{1028}$$

Let y be an atom of F_1. Put

$$a_y = F_1(y-), \qquad b_y = F_1(y).$$

Then $Q_1(u) = y$ for $u \in (a_y, b_y]$. The jump of L^0_{1-} created by this atom has size

$$\Delta L^0_{1-}(y) = \frac{y \int_{a_y}^{b_y} Q_2(1-u)\,du}{J^Q}. \tag{1029}$$

Therefore

$$L^0_{1-} \text{ is continuous} \quad \Longleftrightarrow \quad y \int_{F_1(y-)}^{F_1(y)} Q_2(1-u)\,du = 0 \quad \text{for every atom } y \text{ of } F_1. \tag{1030}$$

Likewise, if z is an atom of F_2, then the jump of L^0_{2-} created by this atom has size

$$\Delta L^0_{2-}(z) = \frac{z\int_{F_2(z-)}^{F_2(z)} Q_1(1-v)\,dv}{J^Q}. \tag{1031}$$

Consequently

$$L^0_{2-} \text{ is continuous} \quad \Longleftrightarrow \quad z\int_{F_2(z-)}^{F_2(z)} Q_1(1-v)\,dv = 0 \quad \text{for every atom } z \text{ of } F_2. \tag{1032}$$

Thus the exact minimal condition on the initial law F, for the *lower-bound* marginals to be atom-free at the initial step, is (1030) and (1032), together with $J^Q > 0$.

A simpler sufficient assumption is

$$F_1 \text{ and } F_2 \text{ are continuous and nondegenerate}, \qquad J^Q > 0. \tag{1033}$$

Here nondegenerate means that the marginal is not a *Dirac mass*. Under (1033), the atom conditions above hold automatically, L^0_{1-} and L^0_{2-} are continuous, and the proof of *Lemma 152* applies as written.

Connection with the Dall'Aglio endpoint cases

We do not repeat here the full Dall'Aglio classification discussed earlier in *Appendix D.4*. We only record the point that is needed for the present appendix.

For $d \geq 3$, the formal *Fréchet–Hoeffding lower-bound* expression

$$F^{\wedge}(x_1, \dots, x_d) = \left(\sum_{i=1}^{d} F_i(x_i) - d + 1\right)_+$$

is a genuine d-dimensional distribution only in exceptional cases. As recalled in *Appendix D.4*, apart from the effectively bivariate case, these are endpoint-jump cases. In the notation used there,

$$q_i = P(X_i > \ell_i), \qquad p_i = P(X_i < r_i),$$

the lower-endpoint case is characterized by

$$\sum_{i=1}^{d} q_i \leq 1,$$

and the upper-endpoint case is characterized by

$$\sum_{i=1}^{d} p_i \leq 1.$$

Thus the endpoint cases are precisely the atomic cases in which the generalized-inverse convention can create rank jumps. The examples below show that these endpoint atoms can lead to denominator collapse.

A bivariate endpoint-atomic lower-bound family

We now give a bivariate example for which the iteration is closed and explicit. Let μ_n denote the probability measure whose distribution function is the n-th *Lorenz iterate*. Fix $a_0 \in (0,1)$, set

$$s_0 = 1 - a_0,$$

and let W_0 be a random variable on $[0,1]$. Define

$$\mu_0 = a_0\delta_{(1,1)} + \frac{s_0}{2}\mathrm{Law}(W_0, 1) + \frac{s_0}{2}\mathrm{Law}(1, W_0). \tag{1034}$$

This is a bivariate *Fréchet–Hoeffding lower-bound* law with endpoint atoms. Indeed, if $p_0 = \frac{s_0}{2}$ and H_0 is the distribution function of W_0, then the common marginal quantile is

$$Q_0(u) = \begin{cases} H_0^{-1}(u/p_0), & 0 < u \le p_0, \\ 1, & p_0 < u \le 1. \end{cases}$$

The countermonotone coupling

$$(Q_0(U), Q_0(1-U)), \qquad U \sim U(0,1),$$

has precisely the law (1034).

Suppose that at step n the law has the form

$$\mu_n = a_n \delta_{(r_n, r_n)} + \frac{s_n}{2}\mathrm{Law}(r_n W_n, r_n) + \frac{s_n}{2}\mathrm{Law}(r_n, r_n W_n), \qquad s_n = 1 - a_n. \tag{1035}$$

Let

$$\theta_n = E[W_n], \qquad D_n = a_n + s_n\theta_n.$$

Then the denominator of the *Lorenz iteration* is

$$I_n = \int x_1 x_2 \, d\mu_n(x_1, x_2) = r_n^2(a_n + s_n\theta_n) = r_n^2 D_n. \tag{1036}$$

The product tilt multiplies the top atom by r_n^2 and each arm by $r_n^2 W_n$. Therefore the new top mass and total arm mass are

$$a_{n+1} = \frac{a_n}{a_n + s_n\theta_n}, \qquad s_{n+1} = \frac{s_n\theta_n}{a_n + s_n\theta_n}. \tag{1037}$$

For each coordinate, the mass below the old upper endpoint r_n is $s_n/2$. Since the generalized-inverse convention sends the upper endpoint atom to the left edge of its jump, the next upper endpoint is

$$r_{n+1} = \frac{s_n}{2}. \tag{1038}$$

It remains to describe the law of the new arm variable. Let H_n be the distribution function of W_n. Let $\widehat{W}_n$ be the size-biased version of W_n:

$$P(\widehat{W}_n \in dw) = \frac{w}{\theta_n} P(W_n \in dw). \tag{1039}$$

On an arm, after product tilting, the varying coordinate is governed by $\widehat{W}_n$. Its old marginal rank inside the arm is $H_n(\widehat{W}_n)$. After division by the new endpoint $r_{n+1} = s_n/2$, the new arm variable is therefore

$$W_{n+1} = H_n(\widehat{W}_n). \tag{1040}$$

Thus the class (1035) is invariant.

The scalar mass ratio has the particularly simple recursion

$$\frac{s_{n+1}}{a_{n+1}} = \theta_n \frac{s_n}{a_n}. \tag{1041}$$

Writing

$$R_n = \frac{s_n}{a_n},$$

we obtain

$$R_n = R_0 \prod_{k=0}^{n-1} \theta_k, \qquad a_n = \frac{1}{1+R_n}, \qquad s_n = \frac{R_n}{1+R_n}. \tag{1042}$$

Moreover,

$$r_{n+1} = \frac{s_n}{2} = \frac{R_n}{2(1+R_n)}. \tag{1043}$$

The beta-arm subclass and explicit collapse

Assume now that

$$W_0 \sim \mathrm{Beta}(\kappa_0, 1), \qquad \kappa_0 > 0.$$

If

$$W_n \sim \mathrm{Beta}(\kappa_n, 1),$$

then

$$H_n(w) = w^{\kappa_n}, \qquad \theta_n = E[W_n] = \frac{\kappa_n}{\kappa_n + 1}.$$

The size-biased version satisfies

$$\widehat{W}_n \sim \mathrm{Beta}(\kappa_n + 1, 1).$$

Hence

$$W_{n+1} = H_n(\widehat{W}_n) = \widehat{W}_n^{\kappa_n} \sim \mathrm{Beta}\left(1 + \frac{1}{\kappa_n}, 1\right).$$

Therefore the beta family is invariant and

$$\kappa_{n+1} = 1 + \frac{1}{\kappa_n}. \tag{1044}$$

The sequence (κ_n) converges to

$$\kappa_* = \frac{1 + \sqrt{5}}{2}.$$

Consequently

$$\theta_n = \frac{\kappa_n}{\kappa_n + 1} \longrightarrow \frac{\kappa_*}{\kappa_* + 1} = \frac{1}{\kappa_*} < 1.$$

Thus the product in (1042) converges to zero exponentially, and

$$R_n \to 0, \qquad s_n \to 0, \qquad a_n \to 1, \qquad r_n \to 0.$$

Using (1036), we conclude

$$I_n = r_n^2 (a_n + s_n \theta_n) \longrightarrow 0. \tag{1045}$$

In the special case $W_0 \sim U(0,1)$, one has $\kappa_0 = 1$, and

$$\kappa_n = \frac{F_{n+2}}{F_{n+1}},$$

where $F_1 = F_2 = 1$ are the *Fibonacci numbers*. Hence

$$\theta_n = \frac{\kappa_n}{\kappa_n + 1} = \frac{F_{n+2}}{F_{n+3}},$$

and therefore

$$R_n = R_0 \prod_{k=0}^{n-1} \frac{F_{k+2}}{F_{k+3}} = \frac{R_0}{F_{n+2}}.$$

Thus, in the uniform-arm case,

$$a_n = \frac{F_{n+2}}{F_{n+2} + R_0}, \qquad s_n = \frac{R_0}{F_{n+2} + R_0}, \qquad r_{n+1} = \frac{R_0}{2(F_{n+2} + R_0)}. \tag{1046}$$

This gives a fully explicit collapsing orbit.

This example shows that the statement

$$\inf_n I_{-,n} > 0$$

is false for arbitrary endpoint-atomic lower-bound trajectories under the literal generalized-inverse convention. The correct statement is that *Lemma 152* proves noncollapse for the atom-free lower-bound trajectory.

The d-dimensional upper-endpoint atomic family

The same mechanism works in all dimensions. Let $d \geq 2$. Fix $a_0 \in (0,1)$, set $s_0 = 1 - a_0$, and let W_0 be a random variable on $[0,1]$. Define

$$\mu_0 = a_0 \delta_{\mathbf{1}} + \frac{s_0}{d} \sum_{i=1}^{d} \mathrm{Law}(1, \ldots, 1, W_0, 1, \ldots, 1), \tag{1047}$$

where W_0 appears in the i-th coordinate and

$$\mathbf{1} = (1, \ldots, 1).$$

For each coordinate,

$$P_{\mu_0}(X_i < 1) = \frac{s_0}{d},$$

and so

$$\sum_{i=1}^{d} P_{\mu_0}(X_i < 1) = s_0 \leq 1.$$

Thus (1047) belongs to the upper-endpoint Dall'Aglio case.

Assume that at step n

$$\mu_n = a_n \delta_{r_n \mathbf{1}} + \frac{s_n}{d} \sum_{i=1}^{d} \mathrm{Law}(r_n, \ldots, r_n, r_n W_n, r_n, \ldots, r_n), \qquad s_n = 1 - a_n. \tag{1048}$$

Let

$$\theta_n = E[W_n], \qquad D_n = a_n + s_n \theta_n.$$

Then

$$I_n = \int \prod_{i=1}^{d} x_i \, d\mu_n(x) = r_n^d (a_n + s_n \theta_n) = r_n^d D_n. \tag{1049}$$

Exactly as in the bivariate case, product tilting gives

$$a_{n+1} = \frac{a_n}{a_n + s_n \theta_n}, \qquad s_{n+1} = \frac{s_n \theta_n}{a_n + s_n \theta_n}. \tag{1050}$$

For each coordinate, the mass below the old upper endpoint r_n is s_n/d. Hence

$$r_{n+1} = \frac{s_n}{d}. \tag{1051}$$

The arm variable evolves by the same size-biased rank recursion:

$$W_{n+1} = H_n(\widehat{W}_n), \qquad P(\widehat{W}_n \in dw) = \frac{w}{\theta_n} P(W_n \in dw). \tag{1052}$$

Therefore

$$\frac{s_{n+1}}{a_{n+1}} = \theta_n \frac{s_n}{a_n}. \tag{1053}$$

If $W_0 \sim \mathrm{Beta}(\kappa_0, 1)$, then again

$$\kappa_{n+1} = 1 + \frac{1}{\kappa_n}.$$

Consequently $s_n/a_n \to 0$, so

$$s_n \to 0, \qquad a_n \to 1, \qquad r_n \to 0.$$

By (1049),

$$I_n = r_n^d (a_n + s_n \theta_n) \longrightarrow 0.$$

Thus the d-dimensional upper-endpoint star family gives a closed-form atomic bad orbit for every $d \geq 2$.

A nonsymmetric atomic family

The symmetry in the preceding example is not essential. Let

$$\mu_n = a_n\delta_{(r_{1,n},\dots,r_{d,n})} + \sum_{i=1}^{d} p_{i,n}\mathrm{Law}(r_{1,n},\dots,r_{i-1,n},r_{i,n}W_{i,n},r_{i+1,n},\dots,r_{d,n}),$$

where

$$a_n + \sum_{i=1}^{d} p_{i,n} = 1.$$

Set

$$\theta_{i,n} = E[W_{i,n}], \qquad D_n = a_n + \sum_{i=1}^{d} p_{i,n}\theta_{i,n}.$$

Then

$$I_n = \left(\prod_{j=1}^{d} r_{j,n}\right) D_n.$$

The mass recursion is

$$a_{n+1} = \frac{a_n}{D_n}, \qquad p_{i,n+1} = \frac{p_{i,n}\theta_{i,n}}{D_n}. \tag{1054}$$

The new upper endpoint in coordinate i is the old mass below the old upper endpoint in that coordinate:

$$r_{i,n+1} = p_{i,n}. \tag{1055}$$

Hence

$$\frac{p_{i,n+1}}{a_{n+1}} = \theta_{i,n}\frac{p_{i,n}}{a_n}. \tag{1056}$$

If each $\theta_{i,n}$ is eventually bounded above by a constant strictly smaller than 1, then

$$p_{i,n} \to 0, \qquad r_{i,n} \to 0, \qquad I_n \to 0.$$

Thus denominator collapse is not an artefact of exchangeability.

Atomic paired countermonotonicity

We next record the corresponding correction for *paired countermonotonicity*. Let $d = 2m$, and pair the coordinates as

$$(1,2),(3,4),\dots,(2m-1,2m).$$

The continuous PCM formulas in the main text remain valid when the block marginals are continuous. With atoms, the block factorization still holds, but the one-dimensional limits in the block integrals must be corrected.

Consider one block (i,j). At step n, let the block marginals be L_i^n, L_j^n, with generalized inverses

$$T_i^n = (L_i^n)^{-1}, \qquad T_j^n = (L_j^n)^{-1}.$$

The countermonotone representation in this block is

$$(X_i, X_j) = (T_i^n(U), T_j^n(1-U)), \qquad U \sim U(0,1).$$

Set

$$\mu_{ij,n} = \int_0^1 T_i^n(u)T_j^n(1-u)\,du.$$

Define the plateau endpoints

$$\alpha_i^n(x) = L_i^n(T_i^n(x)), \qquad \alpha_j^n(x) = L_j^n(T_j^n(x)).$$

Then the corrected atomic block marginal updates are

$$L_i^{n+1}(x) = \frac{\int_0^{\alpha_i^n(x)} T_i^n(u)T_j^n(1-u)\,du}{\mu_{ij,n}}, \tag{1057}$$

and

$$L_j^{n+1}(x) = \frac{\int_{1-\alpha_j^n(x)}^1 T_i^n(u)T_j^n(1-u)\,du}{\mu_{ij,n}}. \tag{1058}$$

Similarly, the corrected bivariate block *Lorenz factor* is

$$L_{ij}^{n+1}(x,y) = \frac{\int_{1-\alpha_j^n(y)}^{\alpha_i^n(x)} T_i^n(u)T_j^n(1-u)\,du}{\mu_{ij,n}}, \tag{1059}$$

with the convention that the numerator is zero if

$$\alpha_i^n(x) < 1 - \alpha_j^n(y).$$

If the block marginals are continuous, then

$$\alpha_i^n(x) = x, \qquad \alpha_j^n(y) = y,$$

and (1057)–(1059) reduce to the continuous PCM formulas.

Finite-support laws and the atomic Lorenz map

We now record the finite-support form of the atomic *Lorenz map*. Let

$$\mu = \sum_{k=1}^N m_k \delta_{x^{(k)}}, \qquad m_k > 0, \qquad \sum_{k=1}^N m_k = 1,$$

where

$$x^{(k)} = (x_1^{(k)}, \dots, x_d^{(k)}) \in [0,1]^d.$$

Assume

$$I(\mu) = \sum_{k=1}^N m_k \prod_{i=1}^d x_i^{(k)} > 0.$$

Define

$$w_k = \prod_{i=1}^d x_i^{(k)}.$$

The product-tilted weights are

$$\widetilde{m}_k = \frac{m_k w_k}{I(\mu)}.$$

Under the generalized-inverse convention, the rank image of the support point $x^{(k)}$ is

$$\rho^{(k)} = (\rho_1^{(k)}, \dots, \rho_d^{(k)}),$$

where

$$\rho_i^{(k)} = \mu\{x : x_i < x_i^{(k)}\}. \tag{1060}$$

Thus the atomic *Lorenz image* is

$$\mathcal{L}(\mu) = \sum_{k=1}^N \widetilde{m}_k \delta_{\rho^{(k)}}, \tag{1061}$$

with atoms having the same rank vector merged. Formula (1061) is the finite-dimensional version of (1025).

Finite-support fixed points and boundary attractors

Suppose that

$$\mu = \sum_{k=1}^{N} m_k \delta_{x^{(k)}}$$

is an atomic fixed point and that no merging of rank images occurs. Then there exists a permutation π of $\{1, \dots, N\}$ such that

$$x^{(\pi(k))} = \rho^{(k)}$$

and

$$m_{\pi(k)} = \frac{m_k w_k}{I(\mu)}, \qquad w_k = \prod_{i=1}^{d} x_i^{(k)}. \tag{1062}$$

Following a cycle

$$k, \pi(k), \dots, \pi^{\ell-1}(k),$$

we obtain

$$I(\mu)^{\ell} = \prod_{r=0}^{\ell-1} w_{\pi^r(k)}. \tag{1063}$$

Thus every cycle of the rank permutation must have product-weight geometric mean equal to the same normalizer $I(\mu)$.

In the special case where the rank image fixes each atom individually,

$$\pi(k) = k,$$

condition (1062) gives

$$w_k = I(\mu) \qquad \text{for every support point } x^{(k)}.$$

Therefore every atom must lie on the same product-level surface

$$\prod_{i=1}^{d} x_i = I(\mu).$$

This is a strong restriction.

The endpoint-star examples above do not converge to genuine interior fixed points. For the bivariate family,

$$\mu_n = a_n \delta_{(r_n, r_n)} + \frac{s_n}{2} \mathrm{Law}(r_n W_n, r_n) + \frac{s_n}{2} \mathrm{Law}(r_n, r_n W_n),$$

the beta-arm computation gives

$$a_n \to 1, \qquad s_n \to 0, \qquad r_n \to 0.$$

Hence

$$\mu_n \Rightarrow \delta_{(0,0)}.$$

But

$$\int x_1 x_2 \, d\delta_{(0,0)} = 0.$$

Thus the limit is not a fixed point of the *Lorenz map*; it is a boundary attractor where the next normalizing denominator is zero. The same conclusion holds in the d-dimensional star family:

$$\mu_n \Rightarrow \delta_{\mathbf{0}}, \qquad I_n \to 0.$$

Discrete complete mixability

Atomic examples also arise from discrete complete mixability. Suppose that at step n the support is indexed by $r = 1, \dots, N$, with masses $m_r^{(n)}$, and that the coordinates are arranged by permutations $\sigma_i \in S_N$:

$$x^{(r,n)} = (x^{(n)}_{1,\sigma_1(r)}, \dots, x^{(n)}_{d,\sigma_d(r)}).$$

A discrete complete mix satisfies

$$\sum_{i=1}^{d} x^{(n)}_{i,\sigma_i(r)} = k_n, \qquad r = 1, \dots, N. \tag{1064}$$

The *Lorenz denominator* is

$$I_n = \sum_{r=1}^{N} m_r^{(n)} \prod_{i=1}^{d} x^{(n)}_{i,\sigma_i(r)}.$$

The tilted masses are

$$\widetilde{m}_r^{(n)} = \frac{m_r^{(n)} \prod_{i=1}^{d} x^{(n)}_{i,\sigma_i(r)}}{I_n}. \tag{1065}$$

The next support is obtained by the left-rank map:

$$x^{(n+1)}_{i,\sigma_i(r)} = \mu_n\{x : x_i < x^{(n)}_{i,\sigma_i(r)}\}, \qquad i = 1, \dots, d. \tag{1066}$$

Atoms with identical rank vectors are merged.

The complete-mixability constraint (1064) is not automatically invariant under the coordinatewise rank map (1066). Therefore a closed all-n discrete complete-mixability theory requires an invariant subclass: after the tilt and rank update, the new support must again be representable by permutations satisfying a complete-mixability identity. In such subclasses, the iteration is finite-dimensional and explicitly computable from (1065)–(1066).

Atomic one-factor Σ-countermonotonicity

Finally, consider an atomic or one-factor model

$$X_i = q_i(U), \qquad i = 1, \dots, d, \qquad U \sim U(0,1),$$

and put

$$K(u) = \prod_{i=1}^{d} q_i(u), \qquad I = \int_0^1 K(u)\, du.$$

The one-step *Lorenz marginal* is exactly

$$L_i(x) = \frac{\int_0^1 K(u) 1_{\{q_i(u) \le T_i(x)\}}\, du}{I}, \tag{1067}$$

where T_i is the marginal generalized inverse.

If $q_i = T_i \circ \pi_i$ for a measure-preserving map π_i, then in the atom-free case one may simplify

$$\{q_i(u) \le T_i(x)\} = \{\pi_i(u) \le x\} \quad \text{up to null sets.}$$

In the atomic case this simplification is not valid without a jump-splitting convention. The correct formula is the event formula (1067), or, in finite support, the left-rank map (1060).

Suppose the one-factor model is Σ-*countermonotone* in the sense that for every nonempty proper subset $A \subset \{1, \dots, d\}$ there is a decreasing function φ_A such that

$$\sum_{j \notin A} q_j(u) = \varphi_A\left(\sum_{i \in A} q_i(u)\right).$$

Product tilting preserves the support, but the subsequent coordinatewise rank transformation need not preserve these functional relations. Hence, as for discrete complete mixability, a closed all-n atomic Σ-*countermonotone* iteration requires an invariant subclass. Outside such subclasses, Σ-*countermonotonicity* gives correct one-step formulas but does not by itself close the *Lorenz dynamics*.

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